

PL/SQL Course Outline

Module 1: Introduction to PL/SQL (1 Lecture)

Topics

- What is PL/SQL?
- Features of PL/SQL
- Advantages over SQL
- Architecture of Oracle Database
- SQL vs PL/SQL
- Structure of a PL/SQL Block

Practical

- First PL/SQL Program
 - Printing output using DBMS_OUTPUT.PUT_LINE
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Module 2: Variables and Data Types (1 Lecture)

Topics

- Declaring variables
- Constants
- Scalar Data Types
- %TYPE
- %ROWTYPE
- Variable Scope

Practical

- Variable declaration
 - Accepting values
 - Displaying output
-

Module 3: Operators and Expressions (1 Lecture)

Topics

- Arithmetic Operators
- Relational Operators
- Logical Operators
- String Operators
- Assignment Operator

Practical Programs

- Calculator
 - Area of Circle
 - Student Percentage
-

Module 4: Conditional Statements (2 Lectures)

Topics

- IF Statement
- IF-ELSE
- Nested IF
- ELSIF
- CASE Statement

Practical Programs

- Grade Calculator
 - Even/Odd
 - Largest of Three Numbers
 - Employee Bonus Calculation
-

Module 5: Loops (2 Lectures)

Topics

- Simple LOOP

- WHILE LOOP
- FOR LOOP
- Nested Loops
- EXIT
- CONTINUE

Practical Programs

- Multiplication Table
 - Factorial
 - Fibonacci Series
 - Prime Number
 - Reverse Number
-

Module 6: Exception Handling (2 Lectures)

Topics

- Types of Exceptions
- Predefined Exceptions
- User-defined Exceptions
- RAISE Statement
- RAISE_APPLICATION_ERROR

Practical Programs

- Divide by Zero
 - NO_DATA_FOUND
 - TOO_MANY_ROWS
 - Invalid Employee ID
-

Module 7: Cursors (2 Lectures)

Topics

- What is Cursor?
- Implicit Cursor
- Explicit Cursor
- Cursor Attributes
- Cursor FOR Loop
- Parameterized Cursor

Practical Programs

- Employee Salary Report
 - Department-wise Employee List
 - Display Top Employees
-

Module 8: Stored Procedures (1 Lecture)

Topics

- Creating Procedures
- Calling Procedures
- IN Parameters
- OUT Parameters
- IN OUT Parameters

Practical Programs

- Calculate Salary
 - Student Result
 - Employee Details
-

Module 9: Functions (1 Lecture)

Topics

- Creating Functions

- Returning Values
- Difference between Procedure and Function

Practical Programs

- GST Calculator
 - Age Calculator
 - Factorial Function
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Module 10: Packages (1 Lecture)**Topics**

- Package Specification
- Package Body
- Advantages of Packages

Practical

- Employee Package
 - Student Package
-

Module 11: Triggers (2 Lectures)**Topics**

- What is Trigger?
- BEFORE Trigger
- AFTER Trigger
- Row-Level Trigger
- Statement-Level Trigger
- INSTEAD OF Trigger

Practical Programs

- Audit Employee Table
 - Prevent Salary Reduction
 - Log Deleted Records
 - Maintain History Table
-

Module 12: Collections (Advanced) (1 Lecture)**Topics**

- Associative Arrays
- Nested Tables
- VARRAY

Practical

- Store Student Marks
 - Employee List
-

Module 13: Dynamic SQL (Advanced)**Topics**

- EXECUTE IMMEDIATE
- Dynamic Queries

Practical

- Create Table Dynamically
 - Dynamic Insert
-

Database for Laboratory

Create tables like:

- EMPLOYEE
- STUDENT
- DEPARTMENT
- COURSE
- CUSTOMER

- ORDERS
- PRODUCT

These tables can be reused throughout the semester.

Mini Project Ideas

Project 1

Employee Payroll Management

Features:

- Add Employee
- Update Salary
- Generate Payslip
- Department Report

Project 2

Student Result Management

Features:

- Add Student
- Enter Marks
- Calculate Percentage
- Grade Calculation
- Result Report

Project 3

Library Management

Features:

- Issue Books
- Return Books
- Fine Calculation
- Book Availability

Laboratory Exercises (20+)

1. Hello PL/SQL
2. Variable Declaration
3. IF Statement
4. IF ELSE
5. Nested IF
6. CASE Statement
7. FOR Loop
8. WHILE Loop
9. Simple LOOP
10. Exception Handling
11. Implicit Cursor
12. Explicit Cursor
13. Cursor FOR Loop
14. Stored Procedure
15. Function
16. Package
17. Trigger
18. BEFORE Trigger
19. AFTER Trigger
20. Dynamic SQL